# MONUM HASHMI

*AI Research Engineer: AI Safety, Security Evaluation & Interpretability*

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## PROFESSIONAL SUMMARY

AI research engineer whose work centers on stress-testing whether AI systems perform as claimed, through negative controls, adversarial evaluation, and interpretability-driven failure analysis, rather than accepting reported results at face value. Direct experience in LLM security, including prompt-injection and data-exfiltration attack classification, and in deploying production LLM agent systems built on third-party frontier model APIs, giving firsthand insight into the operational stakes of AI governance and model-access decisions. Author of four research projects, including compression-robust deepfake detection with explainability diagnostics. Based in Pakistan, bringing a first-hand, international-builder vantage point on how U.S. AI governance and export-control decisions affect downstream developers and institutions abroad. Completing a BS in Artificial Intelligence (expected 2027).

## EDUCATION

**Bachelor of Science in Artificial Intelligence** | *2023 to 2027 (Expected)*
University of Management and Technology, Lahore, Pakistan, GPA: 3.42 / 4.0
Relevant coursework: Deep Learning, Natural Language Processing, Probability & Statistics, Linear Algebra, Data Structures & Algorithms.

## RESEARCH METHODS

- **Model Security & Red-Teaming:** adversarial dataset curation, prompt-injection and data-exfiltration attack classification, grounded-answer and abstention testing for LLM outputs
- **Evaluation & Validation:** subject-independent and cross-domain cross-validation, robustness testing under distribution shift and compression, calibration and error analysis, Macro-F1, MCC, AUROC, confidence intervals
- **Interpretability:** Grad-CAM, SHAP, saliency and attribution analysis, interpretability-driven failure diagnosis, transparent decision rules over black-box classifiers
- **Experimental Design:** hypothesis framing, negative controls and falsification tests, ablation studies, pre-registered analysis plans
- **Research Practice:** reproducible experiment pipelines, version-controlled research code, technical writing, peer review and critique

## TECHNICAL SKILLS

- **LLMs & Agents:** LangChain, LangGraph, multi-agent orchestration, retrieval-augmented generation (RAG), embeddings, vector databases (ChromaDB), prompt engineering, LLM APIs (OpenAI, Anthropic, OpenRouter)
- **Deep Learning & ML:** PyTorch, HuggingFace Transformers, scikit-learn, timm, CNN architectures, BERT fine-tuning, XGBoost
- **Computer Vision:** YOLOv8, DeepSORT, OpenCV, real-time object detection and tracking
- **Backend & Deployment:** FastAPI, Streamlit, Pydantic, REST APIs, n8n workflow automation, AWS EC2, Netlify
- **Programming:** Python, SQL, Git/GitHub, Linux/Bash

## RESEARCH & PUBLICATIONS

**MSAFE-GC: Compression-Robust and Explainable Deepfake Face Detection** | *Under review, Journal of AI and Digital Forensics Research*
Multi-scale adaptive CNN with Grad-CAM used for artefact localisation and failure analysis, evaluated specifically on the low-quality compression tier of FaceForensics++ where most detectors degrade: 96.30% accuracy, 0.9903 AUC, 97.73% F1.

**A Governance Framework for Generative AI in Saudi Higher Education** | *Study design, with Dr. Tariq Mahmood, UMT*
Policy-oriented study of institutional governance mechanisms for generative AI adoption, based on primary-source regulatory review.

**Validity-Aware Evaluation of EEG-ECG Affective-State Fusion Using Pair-Validity Controls** | *Working paper, co-authored with Dr. Tariq Mahmood, UMT*
Introduces two diagnostics, a Pair-Validity Index and a Label-Matching Inflation Index, that test whether reported multimodal fusion gains survive broken-pairing negative controls (wrong-trial, cross-subject label-matched, cross-subject unrestricted). Classical baselines complete under subject-independent four-fold cross-validation on DREAMER.

**Interpretable Violence Detection: A Transparent Motion-Feature Framework** | *Draft in preparation, co-authored with Daniyal Adeeb*

Replaces black-box video classification with hand-designed motion features and transparent decision rules, and shows that feature behaviour is regime-dependent across visual contexts, a failure mode invisible to aggregate accuracy reporting.

## SELECTED PROJECTS

**Application-Layer Intrusion Detection for LLM Systems (Team)** | *github.com/aiexpert-Azan/Application-Layer-Intrusion-Detection-System*
Fine-tuned BERT for four-class detection of prompt-injection, SQL-injection, and PII-leakage attempts against application inputs, reaching 98.88% test accuracy. Owned adversarial dataset curation, the labelling schema that separates attack classes from benign lookalikes, and model training. Tech: HuggingFace Transformers, BERT, PyTorch.

**Multi-Agent Code Review System** | *github.com/monum-hashmi/code-review-agent*
LangGraph pipeline that routes a pull request through specialised correctness, style, and security agents, with ChromaDB retrieval over the surrounding codebase so each review is grounded in repository context rather than the diff alone. Served through FastAPI and deployed on AWS EC2. Tech: LangGraph, FastAPI, ChromaDB, AWS EC2, OpenRouter.

**DeepShield: Compression-Robust, Explainable Deepfake Detection** | *github.com/monum-hashmi/msafe-gc-deepfake-detection*
Research codebase behind MSAFE-GC. Multi-scale adaptive CNN extending FaceForensics++ training, with Grad-CAM used as a diagnostic for failure analysis and artefact localisation rather than post-hoc decoration. Tech: PyTorch, timm, Grad-CAM.

**Suspicious Vehicle and Person Tracker** | *github.com/monum-hashmi/suspicious-vehicle-tracker*
Real-time multi-object tracking with rule-based alerting for stationary person-vehicle pairs inside user-defined regions of interest, calibrated and evaluated against MOT17 ground-truth annotations. Tech: YOLOv8, DeepSORT, OpenCV.

## PROFESSIONAL EXPERIENCE

**AI Engineer, 2M Innovations, Lahore, Pakistan** | *2026 to Present*

- Own client-facing LLM agent systems end to end, retrieval design, prompt architecture, evaluation, and deployment, built on third-party frontier model APIs (OpenAI, Anthropic, OpenRouter) and now handling live customer conversations in production for restaurant and retail clients, replacing manual response handling.
- Reduced unsupported model outputs by separating deterministic transaction logic from LLM-generated answers, and gated every release on grounded-answer checks and adversarial input testing so failure modes surface in review rather than in front of customers.

## CERTIFICATIONS

- Agent Skills with Anthropic, DeepLearning.AI
- AWS Academy Cloud Foundations, Amazon Web Services (2026)
- ChatGPT Prompt Engineering for Developers, DeepLearning.AI

## LANGUAGES

English (Fluent) | Urdu (Fluent)